Discrete Structures

Practice Problems: Generating Functions

Notation. For a sequence $a_0, a_1, a_2, \ldots$ its generating function is

$$A(x) = \sum_{n \geq 0} a_n x^n,$$

and $[x^n]\,A(x)$ denotes the coefficient of x^n in $A(x)$. Problems are arranged in increasing order of difficulty.

1. Find a closed form for the generating function of each sequence.

(a) $0, 0, 0, 1, 1, 1, 1, \ldots$ (b) $1, -1, 1, -1, 1, \ldots$ (c) $1, 3, 9, 27, 81, \ldots$

(d) $2, 0, 2, 0, 2, 0, \ldots$ (e) $\binom{6}{0}, \binom{6}{1}, \ldots, \binom{6}{6}, 0, 0, \ldots$ (f) $1, 2, 3, 4, 5, \ldots$

2. For each generating function, find a formula for a_n, the coefficient of x^n.

(a) $\dfrac{1}{1-5x}$ (b) $\dfrac{x^2}{1-x}$ (c) $(1+x)^4 + \dfrac{1}{1-x}$

(d) $\dfrac{1}{1+2x}$ (e) $\dfrac{1-x^5}{1-x}$ (f) $\dfrac{1}{1-x^3}$

3. Three fair six-sided dice (red, blue, green) are rolled. Write down a generating function whose coefficient of x^n is the number of outcomes with total n. Use it to find the number of outcomes with total 10.

Hint: $1 + x + \cdots + x^m = \dfrac{1 - x^{m+1}}{1 - x}$.

4. Use generating functions to find the number of integer solutions of

$$x_1 + x_2 + x_3 = 12, \qquad 1 \leq x_1 \leq 4, \quad 2 \leq x_2 \leq 5, \quad 3 \leq x_3 \leq 6.$$

Hint: first pull the smallest allowed power out of each bracket.

5. You have unlimited coins of value 1, 2 and 5 rupees. Write the generating function for the number of ways to pay exactly n rupees (order of coins does not matter), and find the number of ways to pay 10 rupees.

6. A shop sells 4 kinds of donuts. In how many ways can you choose 12 donuts if you must take at least 2 of each kind? (Donuts of the same kind are identical.) Answer using a generating function.

Hint: what is the bracket for one kind of donut if you must take at least 2 of it?

7. Use generating functions to solve $a_n = 3a_{n-1} + 2$ for $n \geq 1$, with $a_0 = 1$.

Hint: the constant 2 appears for every $n \geq 1$; what series does it contribute?

8. Use generating functions to solve $a_n = 5a_{n-1} - 6a_{n-2}$ for $n \geq 2$, with $a_0 = 1$ and $a_1 = 4$.

9. A $1 \times n$ strip is tiled with red 1×1 squares, blue 1×1 squares and (uncoloured) 1×2 dominoes. Let t_n be the number of tilings ($t_0 = 1$).

(a) Find the generating function $T(x) = \sum t_n x^n$, without first writing a recurrence.

Hint: compare with how the Fibonacci GF can be written as $1 + (x + x^2) + (x + x^2)^2 + \cdots$.

(b) Find a closed formula for t_n.

10. Use generating functions to solve $a_n = 4a_{n-1} - 4a_{n-2}$ for $n \geq 2$, with $a_0 = 1$ and $a_1 = 6$. (Watch out: the denominator has a repeated factor.)

Hint: recall the series of $\dfrac{1}{(1-x)^2}$.

11. Find $1^2 + 2^2 + \cdots + n^2$ using generating functions.

(a) Starting from $\dfrac{1}{1-x} = \sum_{n\geq 0} x^n$ and differentiating, show that $\displaystyle\sum_{n\geq 0} n^2 x^n = \frac{x(1+x)}{(1-x)^3}$.

(b) Show that if $A(x) = \sum a_n x^n$, then $\dfrac{A(x)}{1-x}$ is the generating function of the partial sums $a_0 + a_1 + \cdots + a_n$.

(c) Deduce a closed formula for $1^2 + 2^2 + \cdots + n^2$.

Hint: $[x^m] \dfrac{1}{(1-x)^k} = \dbinom{m+k-1}{k-1}$.

12. Recall Newton's generalized binomial coefficient $\binom{r}{k} = \dfrac{r(r-1)\cdots(r-k+1)}{k!}$ for any number r and $k \in \mathbb{N}$, and $(1+x)^r = \sum_{k\geq 0} \binom{r}{k} x^k$.

(a) Show that $(1-4x)^{-1/2} = \displaystyle\sum_{n\geq 0} \binom{2n}{n} x^n$.

Hint: write out $\binom{-1/2}{n}$ as a product, and use $1 \cdot 3 \cdot 5 \cdots (2n-1) = \dfrac{(2n)!}{2^n\, n!}$.

(b) Deduce that for every $n \geq 0$, $\displaystyle\sum_{k=0}^{n} \binom{2k}{k}\binom{2n-2k}{n-k} = 4^n$.

Hint: what is the square of the series in (a)?

Optional challenge. Show that the number of ways to write n as a sum of *distinct* positive integers equals the number of ways to write n as a sum of *odd* positive integers (order does not matter). Check it for $n = 6$ first.